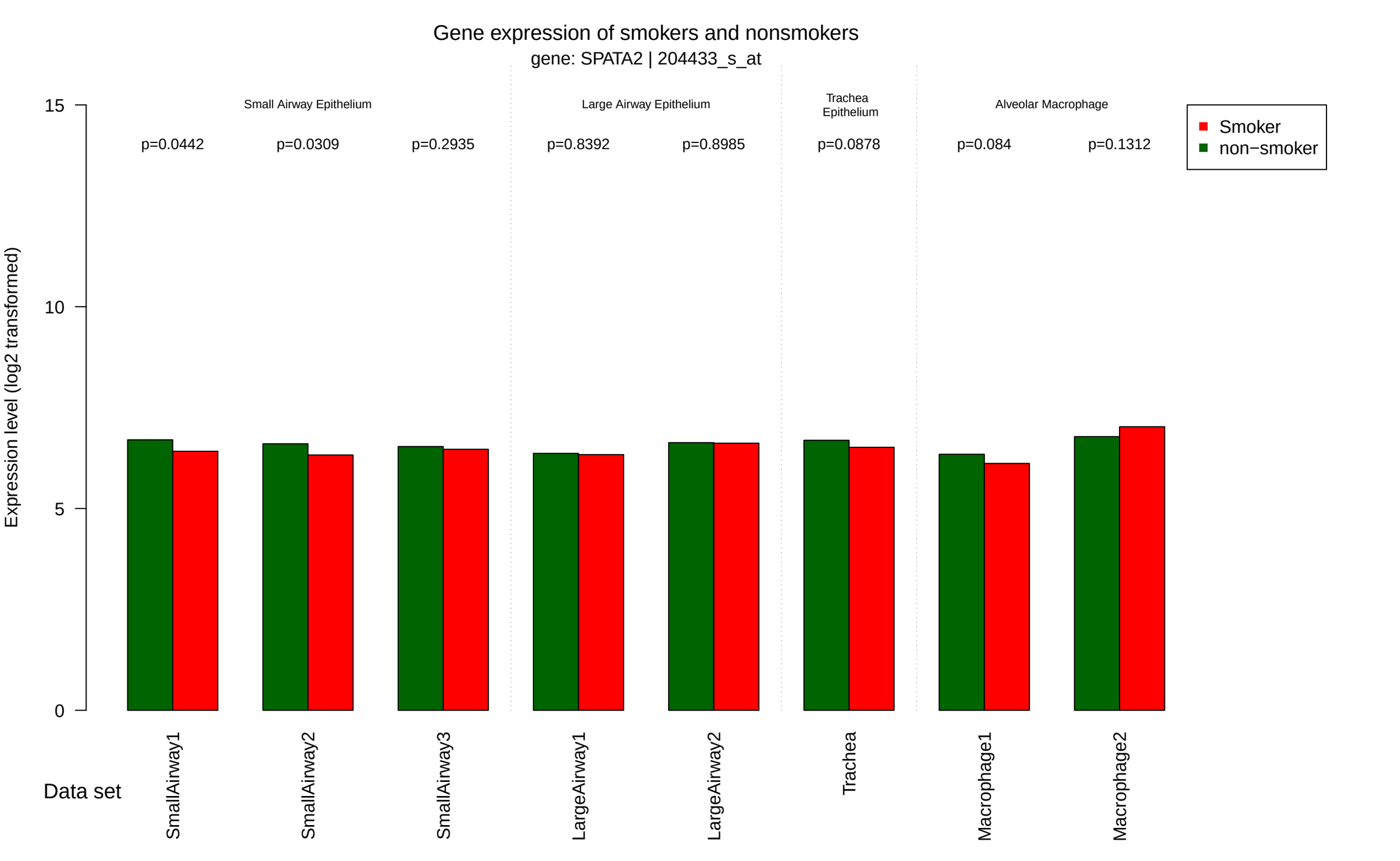
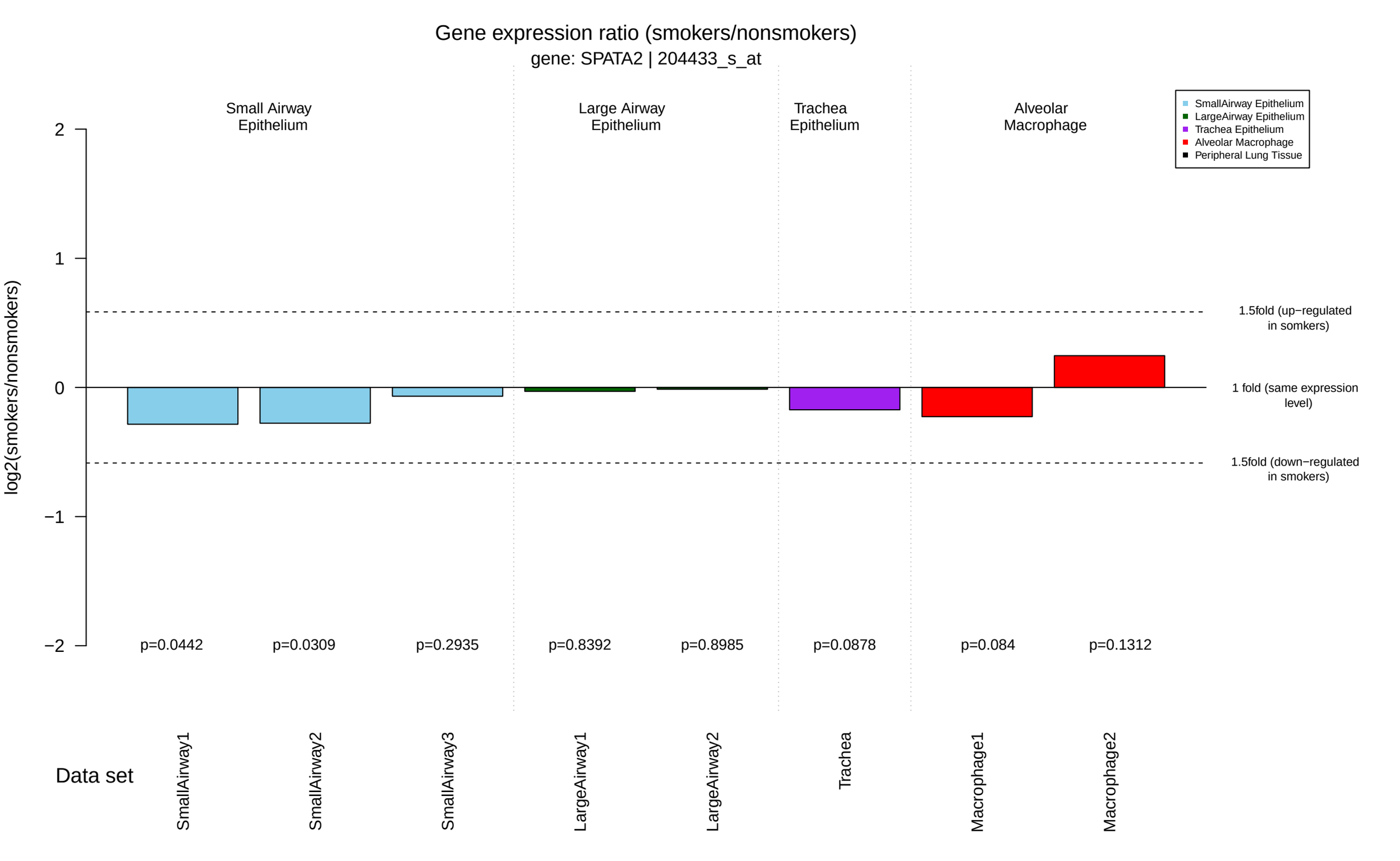
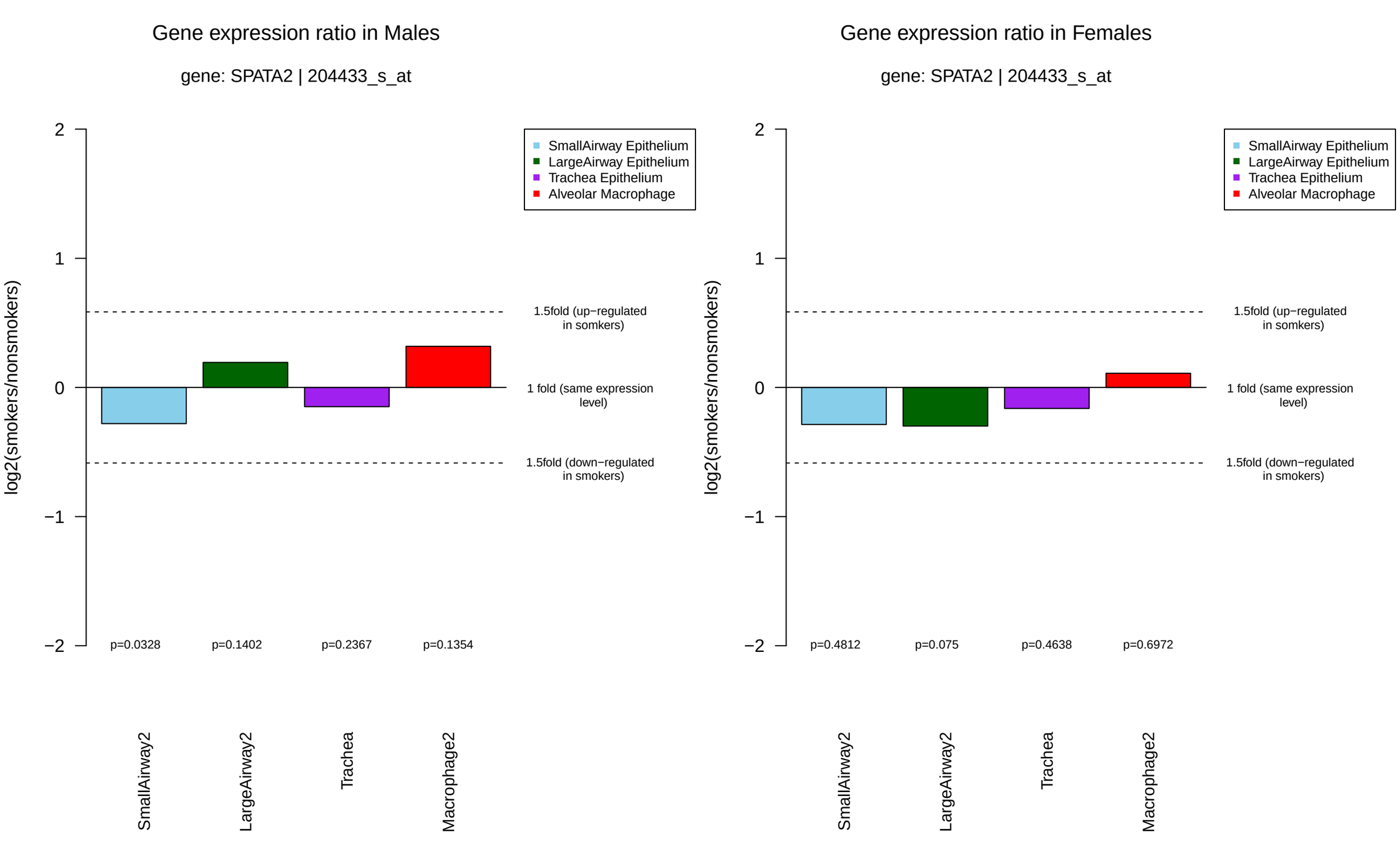
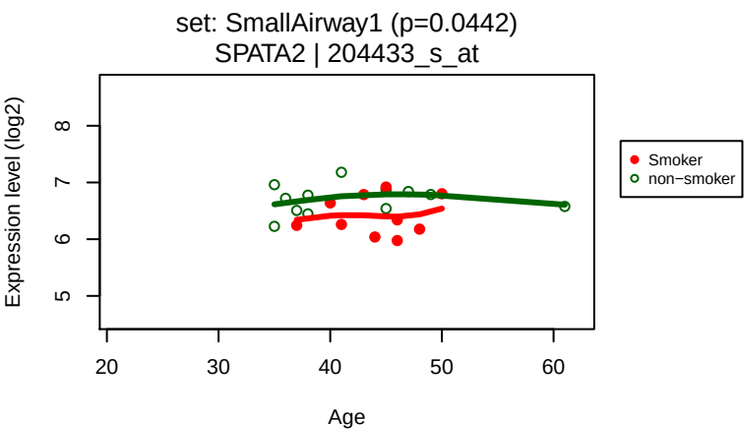
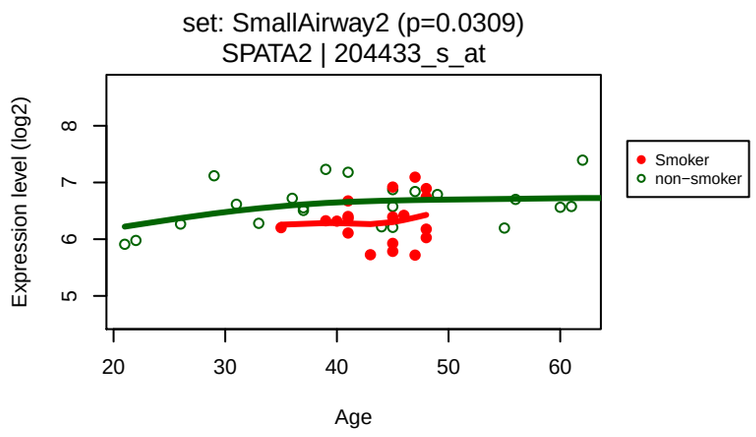
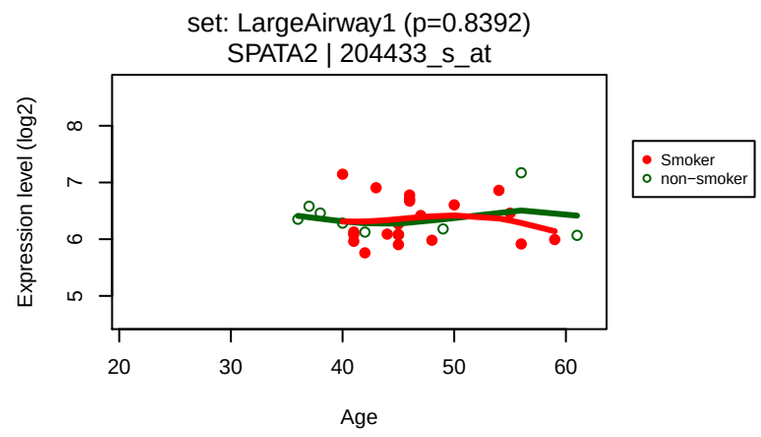

Gene expression of smokers and nonsmokers
gene: SPATA2 | 204433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0442
p=0.0309
p=0.2935
p=0.8392
p=0.8985
p=0.0878
p=0.084
p=0.1312
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

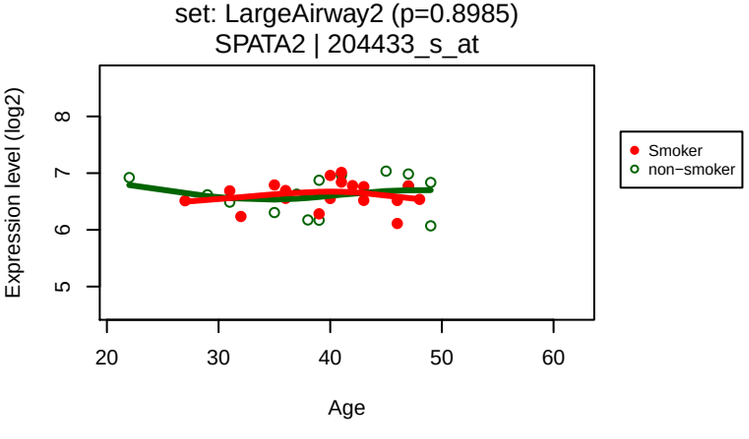

Gene expression ratio (smokers/nonsmokers)
gene: SPATA2 | 204433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0442
p=0.0309
p=0.2935
p=0.8392
p=0.8985
p=0.0878
p=0.084
p=0.1312
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

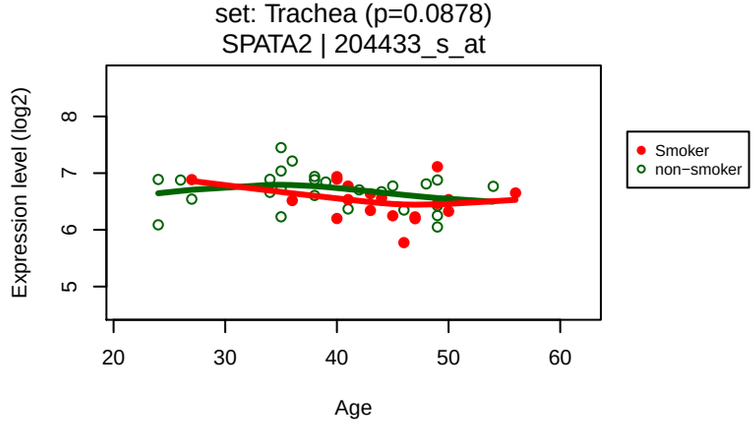

Gene expression ratio in Males
gene: SPATA2 | 204433_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0328
p=0.1402
p=0.2367
p=0.1354
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA2 | 204433_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4812
p=0.075
p=0.4638
p=0.6972
SmallAirway2
LargeAirway2
Trachea
Macrophage2

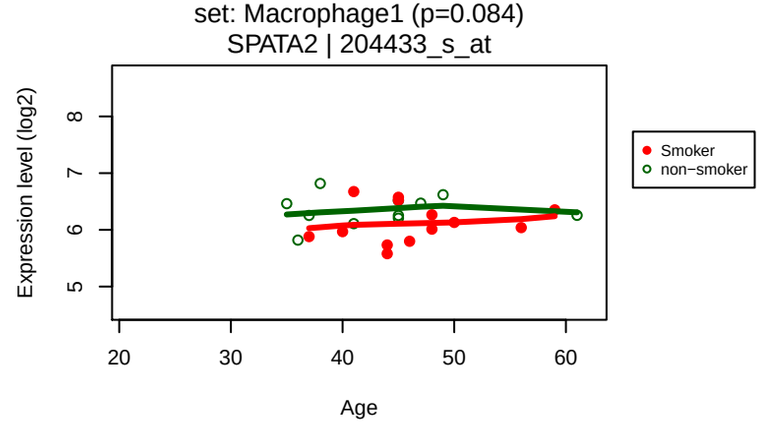

set: SmallAirway1 (p=0.0442)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker

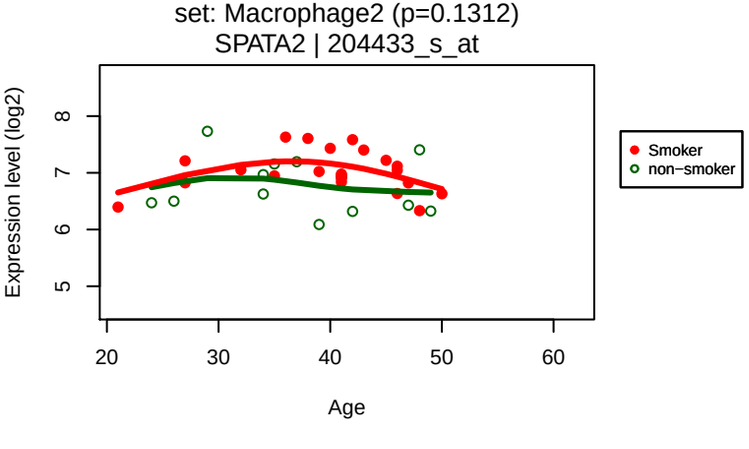

set: SmallAirway2 (p=0.0309)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker

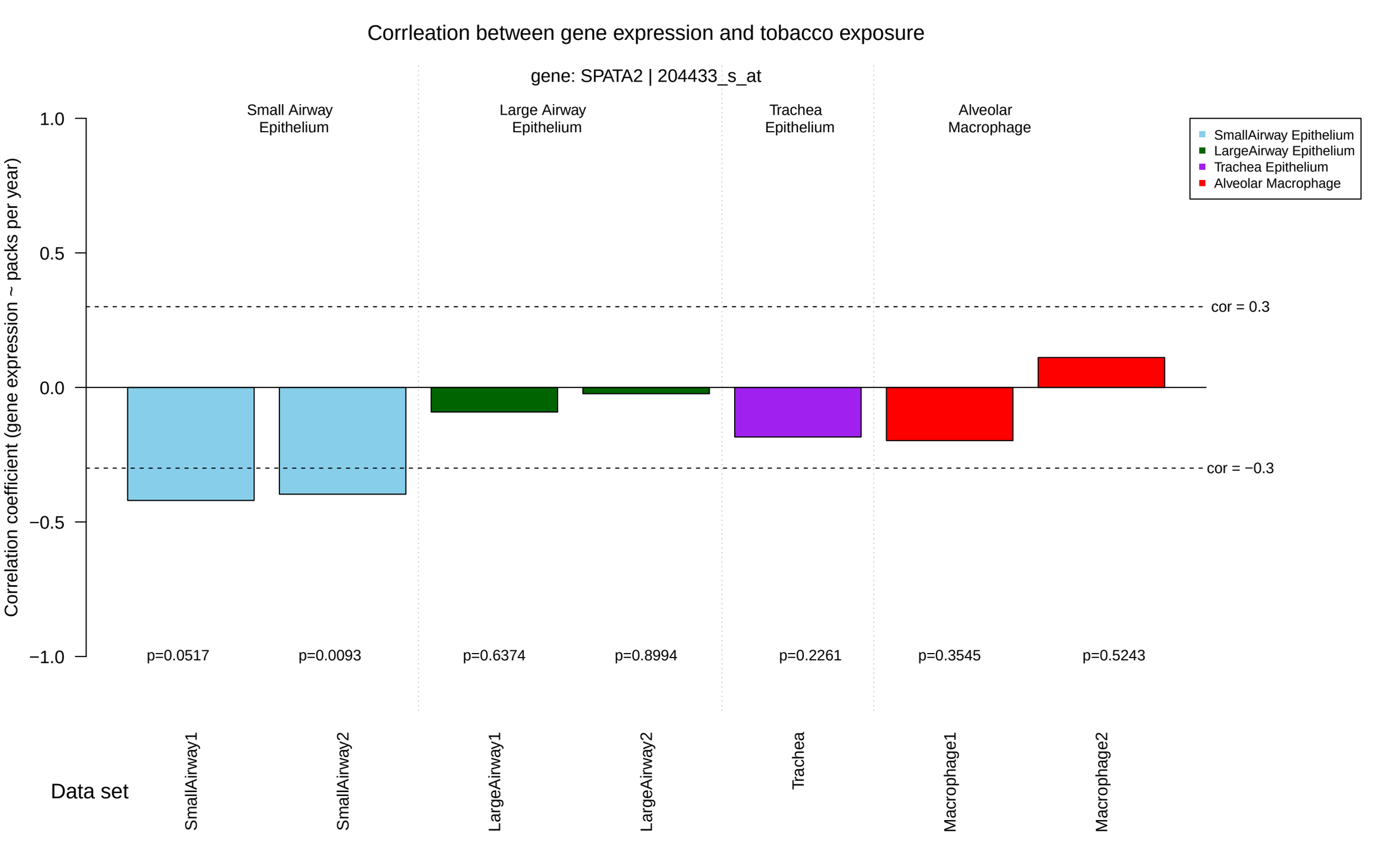

set: LargeAirway1 (p=0.8392)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8985)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0878)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.084)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1312)
SPATA2 | 204433_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA2 | 204433_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0517
p=0.0093
p=0.6374
p=0.8994
p=0.2261
p=0.3545
p=0.5243
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2